

Full Team Project Plan

Topic: Network Reliability under Edge Failures using Monte Carlo Simulation

Project Goal:

To mathematically analyze how cut-edges and minimum cuts determine network reliability under random edge failures, and to experimentally verify these results using Monte Carlo simulation.

Team Roles:

You (Computational Lead):

- Python coding
- Monte Carlo simulation
- Graph generation
- Running experiments
- Plotting results

Math Teammate (Theory & Proof Lead):

- Graph theory definitions
- Mathematical proofs
- Structural analysis- Diagrams and explanations

Project Structure:

1. Introduction and Motivation
 2. Mathematical Theory and Proofs
 3. Monte Carlo Simulation and Experiments
 4. Results, Comparison, and Conclusion
- Detailed Work Breakdown:

Part A – Introduction (Both)

Explain network reliability, real-world motivation, and why Monte Carlo methods are needed.

Part B – Theory and Proofs (Math Teammate)

Cover:

- Connectivity
- Cut-edges (bridges) - Cuts and minimum cut

Key Proofs:

1. A cut-edge failure disconnects the graph.
2. If a graph has k bridges, reliability is at most $(1-q)^k$.
3. If λ is the minimum cut size, disconnection probability is proportional to q^λ .

Part C – Monte Carlo Simulation (You)

Implement:

- Random edge failure simulation
- Connectivity testing- Reliability estimation

Experiments:

- Tree graph - Most Fragile; any edge failure disconnects it.
- Cycle graph - Moderate; requires at least two failures to disconnect.
- Complete graph - Most Robust; requires $n-1$ failures to isolate a node.

Plots:

- Reliability vs probability
- Reliability vs number of bridges
- Reliability vs minimum cut

Part D – Theory vs Simulation Comparison (Both)

Compare theoretical predictions with simulation results.

Report Structure:

1. Introduction
2. Graph Theory Background
3. Mathematical Analysis
4. Monte Carlo Algorithm
5. Simulation Results
6. Discussion
7. Conclusion

Presentation Plan:

- Motivation: Both
- Theory & Proofs: Math Teammate
- Algorithm & Simulations: You
- Results & Conclusion: Both

Timeline:

Week 1:

- Theory development
- Simulation coding

Week 2:

- Experiments

- Report writing- Presentation preparation

Outcome:

A balanced, high-quality project combining mathematics, algorithms, and simulation.